Koblitz compact-orbit single-target panels: n=71 and n=83 hosted isolated reruns

2026-10-14 • run ids `gha-38585913645-1` (n=71), `gha-38597929540-1` (n=83) • status `PENDING_INDEPENDENT_VALIDATION`

What ran

Independent-host reruns of the `koblitz.compact_orbit.n71_single_target` (n=71, a=0) and `koblitz.compact_orbit.n83_single_target` (n=83, a=1) autolab beats on hosted GitHub runners (AMD EPYC 7763, `ubuntu-24.04`), under `tools/isolated_bench.py reserve` with producers pinned to the reserved CPU: 16 one-target eval workloads each, fixed K=600, strong-rho rung 3 comparison arm, one process per arm per workload, empty DP table per process. Targets are public hash-to-curve points with no known scalar; verification is cross-arm scalar agreement plus out-of-process Python certificate replay.

Both runs executed with the point-encoding normalization of PR 1827 without it the local runs of the same beats finished `VERIFICATION_FAILURE` with 16/16 identical scalars (rho's `published_q` decimal strings compared against Python-int targets).

Measured values (hosted, isolated, 0 contended samples)

Beat	K	W	online speedup rho/IC (median, 95% CI)	cold IC/rho	IC / rho online median ms
n=71 a=0	600	16	**7.29x** (4.64–13.38)	154.4	518.6 / 3764.5
n=83 a=1	600	16	**12.29x** (3.26–32.21)	133.7	557.5 / 5442.8

`verify` PASS (256 files, 0 mismatches each). Claim-checks 16/16 PASS per run. Local contended Apple M4 Pro comparison runs read 5.6x (n=71) and 9.46x (n=83) — agreement within the bootstrap intervals, but wall numbers from the contended host are secondary.

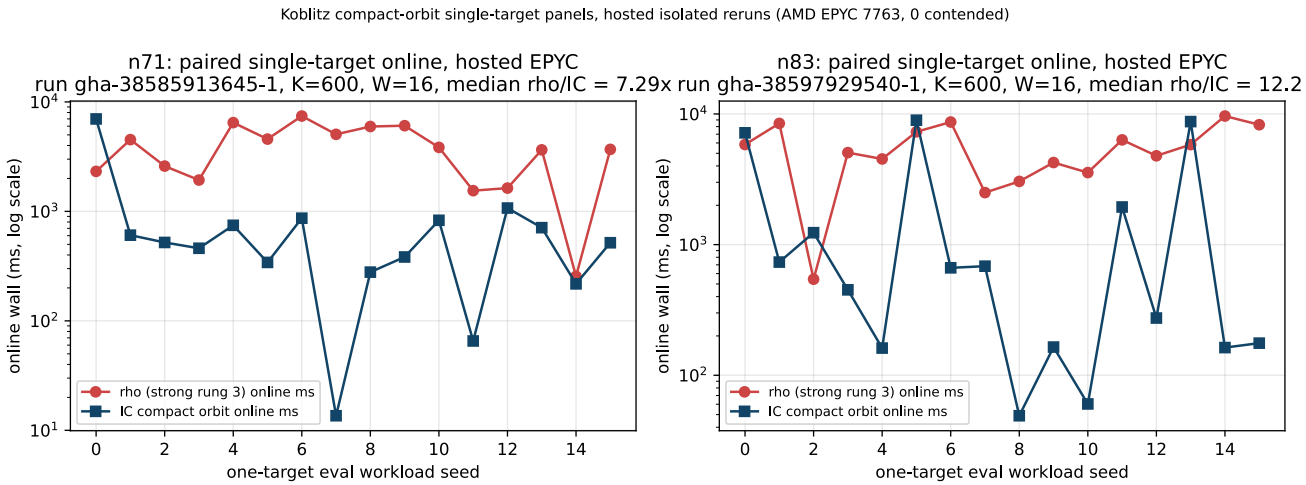


Figure 1: Per-workload paired online walls (log scale) for both hosted runs, recorded values only.

Scope and non-claims

- Public synthetic hash-to-curve targets only; no key recovery, no asymptotic sub-rho claim.
- The IC's reusable setup (500 s in process, mostly rank PDP) is outside the online interval; the cold IC/rho ratio is 154x (n=71) and 134x (n=83) with setup paid per process.
- One host class per arm of the comparison; no claim for other hardware classes.
- Out-of-process replay on the same runner; independent-author replay and a Bernstein–Lange reference remain outstanding. No ledger promotion.

Raw records: `autolab_runs/gha-38585913645-1/`, `autolab_runs/gha-38597929540-1/`, `verify.json`, `isolation.jsonl`, `lscpu.txt` beside this report.